SCX-Audited Educational Assessment:

Multi-Grader Consensus for Certified Scoring

SCXEducational Assessment: Grader

SCX

June 2026

Abstract

Educational assessment—from classroom grading to high-stakes standardized testing—relies on the fundamental act of assigning scores to student work. Yet the scoring enterprise lacks formal certification: a single grader (teacher, examiner, or automated system) produces a single score, and that score is treated as truth. We apply the SCX (Structured Causal eXamination) auditing framework to educational assessment, treating each grader as an **expert** and each score as a **claim** about student competence. The core thesis: unless $M \geq M_{\min}$ independent graders certify a score through YAJIE consensus, that score is unverifiable. We prove five theorems: (1) **Multi-Grader Detection Bound Grader**: Under blind grading conditions (A1 independence), the probability that all M graders miss a scoring error of magnitude Δ is bounded by $\exp(-2M_{\text{eff}}\Delta^2/B^2)$, where B is the scoring range. (2) **Rubric Consensus = Yajie Consensus RubricYajie**: When M graders apply a common rubric $\mathcal{R}$ to student s's work, the YAJIE consensus score $\hat{\theta}_{\text{YAJIE}}$ converges to the true latent score θ_s at rate $O(1/\sqrt{M_{\text{eff}}})$ as grader expertise is bounded below $1/2+\gamma$. (3) **Grade Inflation Detection ScoreInflation**: Systematic upward drift $\iota > 0$ in cohort-averaged scores is detectable with probability $\geq 1 - \exp(-2M_{\text{eff}}\iota^2)$ when M graders are drawn from temporally-independent pools, establishing inflation as a detectable structural bias. (4) **Blind Grading = A1 Enforcement A1**: Removing student identity and inter-grader communication reduces inter-grader error correlation $\bar{\rho}$ by factor $1-\beta$, where $\beta \in [0,1]$ is the blindness index, increasing effective multiplicity $M_{\text{eff}} = M/(1+\bar{\rho}(1-\beta))$. (5) **AI-Human Grading = Multi-Modal Audit AI**: AI graders and human graders constitute heterogeneous expert modalities with distinct error profiles. Their disagreement pattern decomposes into $\text{Var}(\hat{\theta}_{\text{AI}} - \hat{\theta}_{\text{human}}) = \sigma^2_{\text{AI}} + \sigma^2_{\text{human}} - 2\,\text{Cov}(\hat{\theta}_{\text{AI}}, \hat{\theta}_{\text{human}})$, enabling modality-specific calibration. We operationalize these theorems through the CERCIS Score for assessment systems: $S = Q + \eta N$, where Q measures the fraction of SCX theorems the scoring system satisfies and N measures empirical scoring reliability. A single-grader system receives $Q = 0$ and $S \leq 0.20$ regardless of grader expertise — the grader cannot certify their own accuracy (Theorem 2). A K-grader rubric system with blind review receives $Q = 0.60$–0.80. A fully SCX-compliant assessment ($M \geq M_{\min}$ blind graders, SPRING permanent scoring records, YAJIE consensus) achieves $Q = 1.0$ and certified scoring. We provide experimental protocols for the Gaokao (), SAT, and essay-grading contexts, and close with an honest discussion of limitations: the cost of M graders, the impossibility of zero correlation, and the incommensurability of certain rubric dimensions. **Abstract.** Educational Assessment————., : , Score, Score.SCXEducational Assessment, , Score.Core: $M \geq M_{\min}$YAJIEScore, Score.Proof: (1)Grader; (2)RubricYajie; (3)ScoreInflation; (4)A1; (5)AI.CERCIS, , SAT.

Keywords: educational assessment Educational Assessment, multi-grader consensus Grader, grade inflation ScoreInflation, blind grading , standardized testing , AI grading AI, SCX auditing SCX, YAJIE consensus YAJIE, CERCIS score CERCIS, rubric certification Rubric, Gaokao , scoring bias

1 Introduction

Educational assessment is the world's most consequential measurement enterprise. Every year, hundreds of millions of students receive scores — on classroom assignments, end-of-term examinations, university entrance exams (, SAT, A-Levels, Abitur), professional licensing tests, and language proficiency certifications. These scores determine university admissions, scholarship awards, job qualifications, and immigration eligibility. The total stakes, measured in human life trajectories, are incalculable. Yet the scoring enterprise operates under a structural assumption so fundamental that it is rarely examined: **one grader produces one score, and that score is treated as the truth.** Consider three canonical scenarios:

- **Classroom grading .** A teacher grades 30 essays. Each essay receives one score from one grader. If the teacher is tired, biased, or inconsistent — the score stands. There is no second opinion, no audit mechanism, no formal guarantee that the score reflects the student's true competence.
- **Standardized testing .** The Gaokao () processes approximately 12 million scripts annually. Essays are typically double-graded (two human raters), with third-grade arbitration when scores diverge by more than a threshold — an implicit $M = 2$ (or $M = 3$ on disagreement) system. But the graders are trained on the same rubric, calibrated on the same anchor scripts, and subject to the same institutional pressures. Their effective multiplicity M_{eff} may be far lower than the nominal M.
- **AI grading AI.** Automated essay scoring (AES) systems — from e-rater [1] to GPT-4-based graders — produce instantaneous scores at near-zero marginal cost. But an AI grader is $M = 1$: one model, one training corpus, one scoring function. The model cannot certify its own accuracy (Theorem 2 of SCX). When the AI and a human disagree, who is right? Without a formal audit framework, the question is undecidable.

The common thread: **scores are claims about latent student competence, and graders are experts making those claims.** This mapping — graders ↔ experts, scores ↔ claims — is the foundational insight that enables SCX auditing of educational assessment. Every theorem in the SCX framework translates directly to a constraint on scoring validity.

1.1 The Educational Assessment Landscape Educational Assessment

Educational assessment spans a spectrum from low-stakes formative assessment to high-stakes summative certification. We focus on the subset where formal certification matters:

(i) **High-stakes examinations .** Gaokao (, China, ∼12M/year), SAT/ACT (USA, ∼4M/year), GCSE/A-Levels (UK, ∼5M/year), Abitur (Germany), Baccalauréat (France). These examinations determine access to higher education and, by extension, lifetime earnings trajectories.

(ii) **Professional licensing .** Bar examinations, medical board exams (USMLE, PLAB), accounting certifications (CPA, ACCA), engineering licensure (PE, CEng). Scoring errors in these contexts have direct professional consequences.

(iii) **Language proficiency .** IELTS, TOEFL, HSK, DELF/DALF. These determine immigration eligibility and university admission for international students.

(iv) **Large-scale assessments Educational Assessment.** PISA (OECD, ∼80 countries), TIMSS, PIRLS, NAEP (USA). These inform national education policy and international comparisons.

In each context, the assessment system produces scores. The question SCX asks is not "are the scores accurate on average?" — many are. The question is: "**can the assessment system certify the accuracy of each individual score?**"

1.2 The Audit Gap in Education

The audit gap in educational assessment is the distance between *aggregate scoring reliability* (inter-rater reliability coefficients, Cronbach's α, Cohen's κ) and *individual score certification* (knowing whether *this* score for *this* student on *this* question is correct). The assessment literature has developed sophisticated tools for measuring aggregate reliability [2, 3], but these tools do not certify individual scores.

Definition 1.1 (Educational Audit Gap)**.** For an assessment system $\mathcal{A}$ that produces score v_s for student s, the audit gap is:

$$\mathrm{Gap}(\mathcal{A}) = \mathbb{P}(v_s = \theta_s \mid \text{system reports } v_s) - \mathbb{P}(v_s = \theta_s \mid \text{external auditor certifies } v_s), \tag{1}$$

where θ_s is the latent true score. A system with $\mathrm{Gap}(\mathcal{A}) > 0$ has unverifiable scores: the system's self-reported confidence exceeds the confidence warranted by independent audit.

The audit gap manifests in several well-documented phenomena:

(i) **Grade inflation ScoreInflation.** Over decades, mean grades in secondary and tertiary education have drifted upward without corresponding increases in measured competence [4, 6]. In the UK, the proportion of A-level entries awarded A or A* rose from 8.8% (1982) to 44.8% (2021). In US higher education, the modal grade shifted from C (1960s) to A (2010s). A single grader (or institution) cannot detect its own inflation — the drift is visible only through external comparison (Theorem 2: self-audit = no audit).

(ii) **Scoring inconsistency .** The same essay graded by different raters can receive substantially different scores. In a meta-analysis of essay scoring reliability, inter-rater correlations ranged from $r = 0.55$ to $r = 0.85$ [5]. The difference between a passing and failing score can be a matter of which grader was assigned.

(iii) **AI-human disagreement AIDisagreement.** AI grading systems produce scores that correlate with human scores but systematically differ in distribution (lower variance, regression to the mean). Without a multi-expert framework, determining whether the AI or the human is "correct" is logically underdetermined.

(iv) **Rubric ambiguity Rubric.** Scoring rubrics (R) define the mapping from student work to scores. But rubrics are themselves claims about what constitutes quality. A rubric is a consensus among its designers — a Yajie consensus at the design level. When rubrics are applied by a single grader, the consensus is lost.

1.3 Mapping Education to SCX

The conceptual mapping that structures this paper:

1.4 Our Contribution

We provide:

(i) **Formalization** (Section 2): The assessment scoring game with M graders, K rubric dimensions, and explicit grader expertise and bias parameters. Seven assumptions **[A1]**–**[A7]** declared.

(ii) **Five Theorems with Proofs** (Sections 3–7):
- Multi-Grader Detection Bound (Theorem 3.1)
- Rubric Consensus = Yajie Consensus (Theorem 4.1)
- Grade Inflation Detection (Theorem 5.2)

Table 1: Education-to-SCX Mapping

Education Concept	**SCX Concept SCX**	**Justification**
Grader Grader/	Expert	Each grader forms an independent judgment of student competence
Score Score	Claim	A score asserts: "student s's competence is v"
Multi-grader rubric GraderRubric	YAJIE consensus YAJIE	The rubric defines the aggregation rule by which multiple graders' judgments converge
Grade inflation ScoreInflation	Systematic bias	Scores drift upward without corresponding competence improvement — detectable as inter-cohort bias
Blind grading	A1 enforcement A1	Removing student identity and inter-grader communication enforces independent expert training
Standardized test scoring	Multi-expert audit	Multiple graders independently score the same work
AI vs. human grading AI	Multi-modal experts	Different grader modalities (human, AI, rubric-based) with distinct error distributions
Rubric design Rubric	SITUS encoding SITUS	The rubric maps student work (state space) to scores (encoding space) with bounded quality
Score permanence Score	SPRING memory SPRING	Scores and grader identities must be permanently recorded for retrospective audit

- Blind Grading = A1 Enforcement (Theorem 6.2)
- AI-Human Multi-Modal Audit (Theorem 7.2)

(iii) **Cercis Score for Assessment Systems** (Section 8): $S(\mathcal{A}) = Q(\mathcal{A}) + \eta \cdot N(\mathcal{A})$ quantifying the certification quality of any scoring system.

(iv) **Experimental Protocols** (Section 9): Concrete protocols for Gaokao essay scoring, SAT scoring, and AI-assisted grading with Spring permanent records.

(v) **Discussion** (Section 10): Honest limitations — the cost of M graders, irreducible inter-grader correlation, rubric incommensurability, and the political economy of grade inflation.

2 Formalization: The Assessment Scoring Game

2.1 The Assessment State Space

Definition 2.1 (Student Competence). For a student $s \in \mathcal{S}$, the true latent competence on assessment dimension $k \in [K]$ is $\theta_s^{(k)} \in [0, B_k] \subset \mathbb{R}$, where B_k is the maximum score on dimension k. The full competence vector is:

$$\boldsymbol{\theta}_s = (\theta_s^{(1)}, \dots, \theta_s^{(K)}) \in \mathcal{T} = \prod_{k=1}^{K} [0, B_k]. \tag{2}$$

For holistic scoring ($K = 1$), $\theta_s \in [0, B]$.

The competence $\boldsymbol{\theta}_s$ is latent — it cannot be directly observed. Assessment is the process of estimating $\boldsymbol{\theta}_s$ from observable student work w_s (essay, exam answers, project, oral examination).

Definition 2.2 (Student Work). Student s produces work $w_s \in$, where is the space of all possible responses to the assessment task. The mapping from competence to work is stochastic:

$$w_s \sim p(w \mid \boldsymbol{\theta}_s, \xi_s), \tag{3}$$

where ξ_s captures idiosyncratic factors (test anxiety, time pressure, luck in question selection) independent of $\boldsymbol{\theta}_s$.

2.2 The Grader as Expert Grader

Definition 2.3 (Grader Grader). A grader $g \in \mathcal{G} = \{g_1, \dots, g_M\}$ is a function:

$$g_m : \times \mathcal{R} \to [0, B], \tag{4}$$

that maps student work w_s and a rubric $\mathcal{R}$ to a score $v_{s,m} = g_m(w_s, \mathcal{R})$. Each grader has:

- **Expertise** $\mathbf{e}_m \in [0, 1]$: the probability that the grader correctly identifies the quality level of work, $\mathbb{P}(|g_m(w_s, \mathcal{R}) - \theta_s| \le \delta \mid \theta_s) \ge \mathbf{e}_m$ for $\delta > 0$.
- **Bias** $\mathbf{b}_m \in [-B, B]$: systematic over- or under-scoring tendency, $\mathbb{E}[g_m(w_s, \mathcal{R}) - \theta_s] = \mathbf{b}_m$.
- **Variance** σ_m^2: random scoring noise around the biased estimate.

The grader's score production function is:

$$v_{s,m} = \theta_s + \mathbf{b}_m + \varepsilon_{s,m}, \quad \varepsilon_{s,m} \sim \mathcal{D}_m(0, \sigma_m^2), \tag{5}$$

where $\mathcal{D}_m$ is grader m's error distribution (not necessarily Gaussian).

2.3 The Rubric as Situs Encoding RubricSitus

Definition 2.4 (Rubric Rubric)**.** A rubric $\mathcal{R}$ is a Situs encoding that partitions the work space into quality regions and maps each region to a score. Formally:

$$\mathcal{R} = \{\phi_k :\to [0, B_k]\}_{k=1}^{K} \cup \{\psi : [0, B_1] \times \cdots \times [0, B_K] \to\}, \tag{6}$$

where ϕ_k is the scoring function for dimension k and ψ is the (partial) decoder — the exemplar work that "represents" a given score combination. The rubric's quality is:

$$Q_{\text{Situs}}(\mathcal{R}) = (1 - \varepsilon_{\mathcal{R}}) \cdot \exp(-L_{\mathcal{R}} \cdot \sigma_w^2), \tag{7}$$

where $\varepsilon_{\mathcal{R}}$ is the reconstruction error (how well exemplars represent score levels) and $L_{\mathcal{R}}$ is the rubric's Lipschitz constant (sensitivity of scores to small work variations).

A well-designed rubric has low $L_{\mathcal{R}}$ (similar work receives similar scores) and low $\varepsilon_{\mathcal{R}}$ (exemplars accurately represent score levels). A poorly-designed rubric has high $L_{\mathcal{R}}$ (score changes dramatically for minor work differences) or high $\varepsilon_{\mathcal{R}}$ (exemplars are unrepresentative).

2.4 Core Assumptions Core

Assumption 2.5 (Independent Grader Training — A1 Grader)**.** Each grader g_m is trained independently: they receive disjoint calibration sets $\mathcal{D}_m \subset \mathcal{D}_{\text{cal}}$, do not communicate during training, and do not observe other graders' scores before producing their own. Formally, for $m \neq m'$: $I(\mathcal{D}_m; \mathcal{D}_{m'}) = 0$, and grader m's scoring function $g_m(\cdot, \mathcal{R})$ is conditionally independent of $g_{m'}(\cdot, \mathcal{R})$ given $\mathcal{R}$.

Assumption 2.6 (Conditional Grader Independence — A2 Grader)**.** Given the true score θ_s, grader errors are conditionally independent:

$$\mathbb{P}(\varepsilon_{s,m} = a, \varepsilon_{s,m'} = b \mid \theta_s) = \mathbb{P}(\varepsilon_{s,m} = a \mid \theta_s) \cdot \mathbb{P}(\varepsilon_{s,m'} = b \mid \theta_s), \quad \forall m \neq m'. \tag{8}$$

In practice, graders share rubric $\mathcal{R}$, creating correlation. We define the **effective multiplicity**:

$$M_{\text{eff}} = \frac{M}{1 + \bar{\rho}_{\text{grader}}}, \tag{9}$$

where $\bar{\rho}_{\text{grader}} = \frac{2}{M(M-1)} \sum_{m<m'} \text{Corr}(\varepsilon_{s,m}, \varepsilon_{s,m'})$ is the average inter-grader error correlation.

Assumption 2.7 (Bounded Grader Competence — A3 Grader)**.** Each grader achieves accuracy strictly better than random scoring within the score band δ. For B score levels:

$$\mathbb{P}(|v_{s,m} - \theta_s| \leq \delta \mid \theta_s) > \frac{2\delta}{B}, \quad \forall m \in [M]. \tag{10}$$

Equivalently, each grader's scoring function is informative: $\mathbf{e}_m > 2\delta/B$.

Assumption 2.8 (Detectable Scoring Error Margin — A4)**.** When a scoring error of magnitude $\Delta > 0$ occurs, at least one grader detects it with probability $p_d \geq 1/2 + \gamma$ for some $\gamma > 0$:

$$\mathbb{E}[\mathbf{1}[|v_{s,m} - \theta_s| > \Delta] \mid \text{scoring error } \Delta \text{ exists}] \geq \frac{1}{2} + \gamma. \tag{11}$$

Assumption 2.9 (Stationary Student Distribution — A5)**.** The distribution of student competence $\boldsymbol{\theta}_s$ across cohorts is stationary within regime: $\boldsymbol{\theta}_s \sim \mathcal{P}_t$ where $\mathcal{P}_t = \mathcal{P}_{t-1}$ unless a structural break (curriculum reform, demographic shift, assessment redesign) has occurred. Distribution shift $\mathcal{P}_t \to \mathcal{P}_{t+1}$ is detectable via Spring gating.

Assumption 2.10 (Blindness Feasibility — A6). Student identity markers can be removed from work w_s with high probability: $P(\text{identity revealed after redaction}) \leq \varepsilon_{\text{redact}} \ll 1$. For written work, this is achievable through anonymization protocols. For oral examinations or performance assessments, blindness is partial: $\varepsilon_{\text{redact}} > 0$.

Assumption 2.11 (Rubric Stability — A7 Rubric). The rubric $\mathcal{R}$ is stable across the assessment period: $\mathcal{R}_t = \mathcal{R}_{t-1}$ for all t in the assessment window. Changes to $\mathcal{R}$ constitute a structural break and require re-audit.

3 Theorem 1: Multi-Grader Detection Bound

Theorem 3.1 (Multi-Grader Scoring Error Detection Grader). *Let $\mathcal{G} = \{g_1, \ldots, g_M\}$ be M graders satisfying A1–A4, scoring student work on the range $[0, B]$ with effective multiplicity $M_{eff} = M/(1 + \bar{\rho}_{grader})$. For any scoring error of magnitude $\Delta > 0$, the probability that all M graders fail to detect the error satisfies:*

$$\mathbb{P}\left(\bigcap_{m=1}^{M} \{|v_{s,m} - \theta_s| \leq \Delta\}\right) \leq \exp\left(-\frac{2M_{eff}\Delta^2}{B^2}\right). \tag{12}$$

Proof. Define the error detection indicator for grader m:

$$Z_m = \mathbf{1}[|v_{s,m} - \theta_s| > \Delta]. \tag{13}$$

Under A4, $\mathbb{E}[Z_m \mid \text{error exists}] = p_m \geq 1/2 + \gamma$. Under A2, $Z_m \perp Z_{m'}$ given θ_s (with residual correlation $\bar{\rho}_{\text{grader}}$ captured by M_{eff}). The event "all M graders miss the error" is $\bigcap_{m=1}^{M}\{Z_m = 0\}$. By Hoeffding's inequality for the sample mean $\bar{Z} = \frac{1}{M}\sum_{m=1}^{M} Z_m$:

$$\mathbb{P}(\bar{Z} - \mathbb{E}[\bar{Z}] \leq -t) \leq \exp(-2M_{\text{eff}}t^2), \tag{14}$$

where the range of Z_m is $[0, 1]$. Setting $t = \mathbb{E}[\bar{Z}] = \bar{p} \geq 1/2 + \gamma$ and noting that $t \geq \Delta/B$ (since the minimum detectable error as a fraction of the scoring range determines detection probability):

$$\mathbb{P}(\bar{Z} = 0) = \mathbb{P}(\bar{Z} - \bar{p} \leq -\bar{p}) \leq \exp(-2M_{\text{eff}}\bar{p}^2) \leq \exp\left(-\frac{2M_{\text{eff}}\Delta^2}{B^2}\right), \tag{15}$$

where the last inequality uses $\bar{p} \geq \Delta/B$ (a grader detecting error Δ on range B must have accuracy deviating from random by at least Δ/B). □ □

Corollary 3.2 (Single-Grader Insufficiency Grader). *For $M = 1$, the detection probability is at most $1 - \exp(-2\Delta^2/B^2)$. For a typical scoring range $B = 100$ (percentage scale) and minimum meaningful error $\Delta = 10$ (one letter grade), $\mathbb{P}(\text{detect}) \leq 1 - e^{-0.02} \approx 0.0198$. A single grader has less than 2% probability of detecting a 10-point scoring error — the grader is epistemically blind to their own mistakes.*

Corollary 3.3 (Required Number of Graders Grader). *To achieve error detection probability $\geq 1 - \varepsilon$, the required number of independent graders is:*

$$M \geq \frac{B^2 \ln(1/\varepsilon)}{2\Delta^2} \cdot (1 + \bar{\rho}_{grader}). \tag{16}$$

For $B = 100$, $\Delta = 5$, $\varepsilon = 0.05$, $\bar{\rho}_{grader} = 0.3$: $M \geq \frac{10000 \cdot 2.996}{2 \cdot 25} \cdot 1.3 = 77.9 \Rightarrow M \geq 78$. This explains why high-stakes essay scoring with two raters ($M = 2$) provides negligible formal error detection. For $\Delta = 15$ (three letter grades on a 100-point scale), $M \geq 8.7 \Rightarrow M \geq 9$.

Remark 3.4 (The Gaokao Double-Grading Gap). The Gaokao essay scoring protocol uses $M = 2$ graders with third-grade arbitration when scores diverge by τ (typically $\tau = 6$ on a 60-point essay scale). Under our framework, $M = 2$ with $\bar{\rho}_{\text{grader}} \approx 0.5$ (same training, same rubric, shared institutional context) gives $M_{\text{eff}} = 2/1.5 \approx 1.33$. The detection probability for a $\Delta = 10\%$ error on a 60-point scale ($\Delta = 6$) is:

$$\mathbb{P}(\text{detect}) \leq 1 - \exp\left(-\frac{2 \cdot 1.33 \cdot 6^2}{60^2}\right) = 1 - e^{-0.0266} \approx 0.026. \tag{17}$$

The system detects at most 2.6% of significant scoring errors — effectively no detection. The third-grade arbitration improves this slightly: $M_{\text{eff}} \approx (3)/(1+0.5) = 2.0$ when triggered, yielding $\mathbb{P}(\text{detect}) \leq 1 - e^{-0.04} \approx 0.039$.

4 Theorem 2: Rubric Consensus = Yajie Consensus

Theorem 4.1 (Rubric-Guided Yajie Consensus RubricYajie)**.** *Let M graders produce scores $\{v_{s,m}\}_{m=1}^{M}$ for student s under rubric $\mathcal{R}$ with A1–A4. Define the* Yajie *consensus score:*

$$\hat{\theta}_{\text{YAJIE}}(s) = \sum_{m=1}^{M} \omega_m \cdot v_{s,m}, \quad \omega_m = \frac{1/\sigma_m^2}{\sum_{j=1}^{M} 1/\sigma_j^2}, \tag{18}$$

where $\sigma_m^2 = \mathrm{Var}(v_{s,m} \mid \theta_s)$ is grader m's scoring variance. Then:

$$\mathbb{E}\left[(\hat{\theta}_{\text{YAJIE}}(s) - \theta_s)^2\right] \leq \frac{1}{\sum_{m=1}^{M} 1/\sigma_m^2} + \left(\sum_{m=1}^{M} \omega_m \mathbf{b}_m\right)^2, \tag{19}$$

and the consensus converges to the unbiased true score at rate $O(1/\sqrt{M_{\mathit{eff}}})$ when $\mathbf{b}_m = 0$ for all m.

Proof. Decompose the Yajie consensus error:

$$\begin{aligned} \hat{\theta}_{\text{YAJIE}}(s) - \theta_s &= \sum_{m=1}^{M} \omega_m (v_{s,m} - \theta_s) && (20) \\ &= \sum_{m=1}^{M} \omega_m (\mathbf{b}_m + \varepsilon_{s,m}) && (21) \\ &= \underbrace{\sum_{m=1}^{M} \omega_m \mathbf{b}_m}_{\text{systematic bias}} + \underbrace{\sum_{m=1}^{M} \omega_m \varepsilon_{s,m}}_{\text{random error}}. && (22) \end{aligned}$$

By A2 (conditional independence of errors), the variance of the random error term is:

$$\mathrm{Var}\left(\sum_{m=1}^{M} \omega_m \varepsilon_{s,m}\right) = \sum_{m=1}^{M} \omega_m^2 \sigma_m^2. \tag{23}$$

Minimizing this variance subject to $\sum_m \omega_m = 1$ via Lagrange multipliers yields $\omega_m \propto 1/\sigma_m^2$, giving:

$$\mathrm{Var}_{\text{random}} = \frac{1}{\sum_{m=1}^{M} 1/\sigma_m^2}. \tag{24}$$

The total MSE decomposes as variance plus squared bias:

$$\text{MSE}(\hat{\theta}_{\text{YAJIE}}) = \frac{1}{\sum_{m=1}^{M} 1/\sigma_m^2} + \left(\sum_{m=1}^{M} \omega_m \mathbf{b}_m\right)^2. \tag{25}$$

When $\mathbf{b}_m = 0$ for all m and $\sigma_m^2 = \sigma^2$ for all m (homogeneous graders), MSE $= \sigma^2/M$, yielding $O(1/\sqrt{M})$ convergence. With inter-grader correlation $\bar{\rho}$, the effective sample size is $M_{\text{eff}} = M/(1+\bar{\rho})$, giving $O(1/\sqrt{M_{\text{eff}}})$ convergence. □ □

Corollary 4.2 (Rubric Design as Variance Reduction Rubric)**.** *A rubric $\mathcal{R}$ reduces inter-grader variance σ_m^2 by providing explicit scoring anchors. The rubric's effectiveness is:*

$$\eta_{\mathcal{R}} = 1 - \frac{\sigma^2_{\textit{with rubric}}}{\sigma^2_{\textit{without rubric}}}. \tag{26}$$

A rubric with $\eta_{\mathcal{R}} = 0.5$ halves inter-grader variance, equivalent to doubling the effective number of graders without increasing M.

Corollary 4.3 (Score Confidence Interval Score)**.** *Under normally-distributed grader errors, the $(1-\alpha)$ confidence interval for θ_s is:*

$$\hat{\theta}_{\text{YAJIE}}(s) \pm z_{\alpha/2} \cdot \sqrt{\frac{1}{\sum_{m=1}^{M} 1/\hat{\sigma}_m^2} + \widehat{\text{Bias}}^2}, \tag{27}$$

where $\widehat{\text{Bias}} = \sum_m \omega_m \mathbf{b}_m$ is estimated from historical grading data in Spring *memory. A score report should display this interval, not a point estimate.*

5 Theorem 3: Grade Inflation Detection

Grade inflation is the systematic upward drift of scores over time without corresponding improvement in measured competence. It is a canonical example of SCX systematic bias — a structural error that no single grader can detect (Theorem 2) but that multi-grader temporal audit makes visible.

Definition 5.1 (Grade Inflation Rate ScoreInflation)**.** For cohorts $t = 1, 2, \ldots, T$, let $\bar{v}_t = \frac{1}{|\mathcal{S}_t|} \sum_{s \in \mathcal{S}_t} v_s$ be the cohort-averaged score. The grade inflation rate from cohort $t-1$ to t is:

$$\iota_t = \frac{\bar{v}_t - \bar{v}_{t-1}}{\bar{v}_{t-1}}. \tag{28}$$

Inflation is present when $\iota_t > 0$ but the latent competence distribution is stationary: $\mathcal{P}_t = \mathcal{P}_{t-1}$.

Theorem 5.2 (Grade Inflation Detection ScoreInflation)**.** *Let cohorts $t-1$ and t be scored by disjoint grader pools $\mathcal{G}_{t-1}$ and $\mathcal{G}_t$, each of size M, with A1–A5. Define the inter-cohort score drift:*

$$D_t = \bar{v}_t - \bar{v}_{t-1} = \frac{1}{|\mathcal{S}_t|} \sum_{s \in \mathcal{S}_t} v_s - \frac{1}{|\mathcal{S}_{t-1}|} \sum_{s' \in \mathcal{S}_{t-1}} v_{s'}. \tag{29}$$

Under the null hypothesis of no inflation ($\mathcal{P}_t = \mathcal{P}_{t-1}$ and no grader bias shift), D_t has mean zero. The probability of failing to detect genuine inflation of magnitude $\iota > 0$ is:

$$\mathbb{P}(\textit{miss inflation} \mid \iota) \le \exp\left(-\frac{2M_{\textit{eff}}\iota^2 \cdot \bar{v}^2}{B^2}\right). \tag{30}$$

Proof. Under A5 (stationary student distribution), the expected score for any student under unbiased grading is $\mathbb{E}[v_s] = \mathbb{E}_{\theta\sim\mathcal{P}}[\theta] = \mu_\theta$. The observed cohort mean $\bar{v}_t$ estimates μ_θ with two sources of variation: sampling variation across students and grader variation. Decompose $\bar{v}_t$:

$$\bar{v}_t = \underbrace{\frac{1}{|\mathcal{S}_t|}\sum_s \theta_s}_{\text{true mean } \hat{\mu}_t} + \underbrace{\frac{1}{|\mathcal{S}_t|}\sum_s \frac{1}{M}\sum_m (\mathbf{b}_m^{(t)} + \varepsilon_{s,m}^{(t)})}_{\text{grader contribution } \bar{\varepsilon}_t}. \tag{31}$$

With M graders per cohort, by Hoeffding, the grader contribution has variance $\leq B^2/(4M_{\text{eff}})$. For large $|\mathcal{S}_t|$, student sampling variance $\to 0$, and D_t is dominated by grader effects. The inflation signal $\iota \cdot \bar{v}$ is detectable when it exceeds the grader noise floor $B/\sqrt{2M_{\text{eff}}}$:

$$\mathbb{P}(\text{miss}) = \mathbb{P}(|D_t| < \tau \mid \iota) \leq \exp\left(-\frac{2M_{\text{eff}}(\iota\cdot\bar{v} - \tau)^2}{B^2}\right) \leq \exp\left(-\frac{2M_{\text{eff}}\iota^2\bar{v}^2}{B^2}\right), \tag{32}$$

setting the detection threshold $\tau = 0$ for maximum sensitivity. □ □

Corollary 5.3 (Inflation Requires External Audit ScoreInflation)**.** *If the same grader pool $\mathcal{G}$ scores both cohorts $t-1$ and t, the inflation detection bound collapses: grader bias shifts $\Delta\mathbf{b}_m = \mathbf{b}_m^{(t)} - \mathbf{b}_m^{(t-1)}$ are confounded with genuine competence shifts. This is the grade-inflation analog of Theorem 2 (self-audit = no audit): the institution that produced the scores cannot certify that scores haven't inflated. External, temporally-independent grader pools are required.*

Remark 5.4 (UK A-Level and US GPA Inflation A-LevelGPAInflation)*.* The UK A-Level A/A* rate rose from 8.8% (1982) to 44.8% (2021), a compound annual inflation rate of approximately 4.0%. The US college GPA rose from approximately 2.6 (1960s) to 3.15 (2010s), a compound annual inflation rate of approximately 0.4%. Under Theorem 5.2 with $B = 100$, $\bar{v} = 50$, $\iota = 0.04$, and $M_{\text{eff}} = 5$ (typical for institution-level audit): $\mathbb{P}(\text{miss}) \leq \exp(-2\cdot 5\cdot 0.04^2 \cdot 2500/10000) = \exp(-0.04) \approx 0.96$. This means even $M = 5$ independent graders would miss a single year's inflation with 96% probability — but over 39 years (UK) or 50 years (US), the cumulative detection probability approaches 1. The data are clear in hindsight; the framework shows why they were invisible year-by-year.

6 Theorem 4: Blind Grading = A1 Enforcement

Blind grading — the practice of concealing student identity from graders and preventing inter-grader communication — is standard in high-stakes assessment. Our framework reveals *why* it matters mathematically: blindness enforces Assumption A1 (independent expert training) by reducing inter-grader correlation.

Definition 6.1 (Blindness Index)**.** The blindness index $\beta \in [0, 1]$ measures the degree to which student identity and inter-grader influence are suppressed:

- $\beta = 0$: fully non-blind. Graders know student identity and can communicate.
- $\beta = 1$: fully blind. Student identity is perfectly concealed; graders are isolated.

In practice, β is estimated as the reduction in inter-grader correlation attributable to blinding procedures.

Theorem 6.2 (Blindness Increases Effective Multiplicity)**.** *Under blindness level β, the effective multiplicity of M graders is:*

$$M_{\textit{eff}}(\beta) = \frac{M}{1 + \bar{\rho}_{\textit{grader}} \cdot (1-\beta)}, \tag{33}$$

where $\bar{\rho}_{grader}$ is the inter-grader error correlation under $\beta = 0$. Blindness increases detection probability by factor:

$$\frac{\mathbb{P}_\beta(\mathit{detect})}{\mathbb{P}_0(\mathit{detect})} \approx \exp\left(2M\Delta^2 \cdot \frac{\bar{\rho}_{grader} \cdot \beta}{(1+\bar{\rho}_{grader})(1+\bar{\rho}_{grader}(1-\beta))}\right) > 1. \tag{34}$$

Proof. Under non-blind conditions ($\beta = 0$), graders may be influenced by:

(i) **Student identity effects** : Knowing a student's past performance, demographic characteristics, or perceived ability biases scoring. This creates correlation because multiple graders observe the same identity signal.

(ii) **Inter-grader influence Grader**: Graders who discuss scores or observe each other's ratings converge — reducing effective independence.

(iii) **Institutional pressure** : Graders in the same institution share norms about "appropriate" score distributions.

These effects contribute to $\bar{\rho}_{\text{grader}}$. Blinding reduces each: identity effects are eliminated (up to redaction quality $\varepsilon_{\text{redact}}$), inter-grader influence is blocked by isolation protocols, and institutional pressure is weakened when graders don't know whose work they're scoring. Let $\bar{\rho}_{\text{grader}}(\beta) = \bar{\rho}_{\text{grader}}(0) \cdot (1-\beta)$ be the residual correlation under blindness β. Then:

$$M_{\text{eff}}(\beta) = \frac{M}{1 + \bar{\rho}_{\text{grader}}(0) \cdot (1-\beta)}. \tag{35}$$

The detection probability ratio follows from substituting $M_{\text{eff}}(\beta)$ into Theorem 3.1:

$$\frac{\mathbb{P}_\beta(\text{detect})}{\mathbb{P}_0(\text{detect})} = \frac{1 - \exp(-2M_{\text{eff}}(\beta)\Delta^2/B^2)}{1 - \exp(-2M_{\text{eff}}(0)\Delta^2/B^2)} \tag{36}$$

$$\approx \exp\left(2M\Delta^2\left[\frac{1}{1+\bar{\rho}(1-\beta)} - \frac{1}{1+\bar{\rho}}\right]/B^2\right) \tag{37}$$

$$= \exp\left(2M\Delta^2 \cdot \frac{\bar{\rho}\beta}{(1+\bar{\rho})(1+\bar{\rho}(1-\beta))}/B^2\right). \tag{38}$$

Since $\bar{\rho}, \beta \geq 0$, the exponent is non-negative, so the ratio ≥ 1. □ □

Corollary 6.3 (Blindness Multiplier). *For $M = 3$, $\bar{\rho} = 0.5$, complete blindness ($\beta = 1$) transforms effective multiplicity from $M_{\mathit{eff}}(0) = 3/1.5 = 2.0$ to $M_{\mathit{eff}}(1) = 3/1.0 = 3.0$ — a 50% increase. The detection probability for $\Delta/B = 0.1$ (10% of scoring range) improves from $1 - e^{-0.04} \approx 3.9\%$ to $1 - e^{-0.06} \approx 5.8\%$ — a 49% relative improvement. For $M = 10$: from $M_{\mathit{eff}} = 6.67$ to $M_{\mathit{eff}} = 10$ (50% increase), detection probability from $1 - e^{-0.133} \approx 12.5\%$ to $1 - e^{-0.20} \approx 18.1\%$.*

Remark 6.4 (Blindness in Oral Examinations). For oral examinations (PhD defenses, language proficiency interviews, medical OSCEs), perfect blindness ($\beta = 1$) is impossible — the student's identity is visible. However, partial blindness is achievable: multiple examiners can be drawn from different institutions, with no prior knowledge of the student. In this case, β reflects the fraction of identity-correlated information unavailable to graders, and M_{eff} is correspondingly reduced. This explains why doctoral defenses typically use external examiners: to maximize β given the inherent limitations of the format.

7 Theorem 5: AI-Human Grading = Multi-Modal Audit

AI grading systems — from classical AES (e-rater, Intellimetric) to LLM-based graders (GPT-4, Claude) — introduce a new grader modality with distinct error characteristics. The interaction between AI and human graders creates a multi-modal audit structure: two expert types with different training, different biases, and different failure modes.

Definition 7.1 (Multi-Modal Grader Set Grader)**.** A multi-modal grader set consists of M_H human graders $\mathcal{G}_H = \{g_1^H, \ldots, g_{M_H}^H\}$ and M_A AI graders $\mathcal{G}_A = \{g_1^A, \ldots, g_{M_A}^A\}$. Human graders have error distribution $\mathcal{D}_H(0, \sigma_H^2)$ with bias $\mathbf{b}_H$; AI graders have error distribution $\mathcal{D}_A(0, \sigma_A^2)$ with bias $\mathbf{b}_A$. The two modalities are conditionally independent given θ_s: $\mathrm{Cov}(\varepsilon_H, \varepsilon_A \mid \theta_s) = 0$.

Theorem 7.2 (AI-Human Disagreement Decomposition AIDisagreement)**.** *The disagreement between AI and human consensus scores decomposes as:*

$$\mathbb{E}[(\hat{\theta}_{AI} - \hat{\theta}_{human})^2] = \underbrace{(\mathbf{b}_A - \mathbf{b}_H)^2}_{\textit{bias difference}} + \underbrace{\frac{\sigma_A^2}{M_A} + \frac{\sigma_H^2}{M_H}}_{\textit{variance terms}}. \tag{39}$$

The probability that AI and human graders disagree by more than τ while the true score lies between their estimates is bounded by:

$$\mathbb{P}(|\hat{\theta}_{AI} - \hat{\theta}_{human}| > \tau \mid \theta_s \in [\hat{\theta}_{AI}, \hat{\theta}_{human}]) \leq \exp\left(-\frac{\tau^2}{2(\sigma_A^2/M_A + \sigma_H^2/M_H)}\right). \tag{40}$$

Proof. The AI and human consensus scores are:

$$\hat{\theta}_{\mathrm{AI}} = \frac{1}{M_A}\sum_{m=1}^{M_A} v_{s,m}^A, \quad \hat{\theta}_{\mathrm{human}} = \frac{1}{M_H}\sum_{m=1}^{M_H} v_{s,m}^H. \tag{41}$$

By the scoring production function (equation for $v_{s,m}$):

$$\begin{aligned}\hat{\theta}_{\mathrm{AI}} - \hat{\theta}_{\mathrm{human}} &= (\theta_s + \mathbf{b}_A + \bar{\varepsilon}_A) - (\theta_s + \mathbf{b}_H + \bar{\varepsilon}_H) && (42)\\ &= (\mathbf{b}_A - \mathbf{b}_H) + (\bar{\varepsilon}_A - \bar{\varepsilon}_H), && (43)\end{aligned}$$

where $\bar{\varepsilon}_A = \frac{1}{M_A}\sum_m \varepsilon_{s,m}^A$ with variance σ_A^2/M_A, and similarly for humans. Taking expectation of the squared difference:

$$\mathbb{E}[(\hat{\theta}_{\mathrm{AI}} - \hat{\theta}_{\mathrm{human}})^2] = (\mathbf{b}_A - \mathbf{b}_H)^2 + \frac{\sigma_A^2}{M_A} + \frac{\sigma_H^2}{M_H}, \tag{44}$$

since $\mathbb{E}[\bar{\varepsilon}_A] = \mathbb{E}[\bar{\varepsilon}_H] = 0$ and $\mathrm{Cov}(\bar{\varepsilon}_A, \bar{\varepsilon}_H) = 0$ by conditional independence. For the tail bound: when $\mathbf{b}_A = \mathbf{b}_H$ (unbiased graders), the disagreement is $\bar{\varepsilon}_A - \bar{\varepsilon}_H \sim (0, \sigma_A^2/M_A + \sigma_H^2/M_H)$ (asymptotically normal by CLT). By the Gaussian tail bound:

$$\mathbb{P}(|\bar{\varepsilon}_A - \bar{\varepsilon}_H| > \tau) \leq 2\exp\left(-\frac{\tau^2}{2(\sigma_A^2/M_A + \sigma_H^2/M_H)}\right). \tag{45}$$

Conditioning on θ_s lying between the estimates removes the factor of 2 (one tail is impossible), yielding the bound. □ □

Corollary 7.3 (AI Bias Detection AI)**.** *AI grading bias $\mathbf{b}_A$ is detectable through systematic deviation from human consensus:*

$$\hat{\mathbf{b}}_A = \frac{1}{|\mathcal{S}|}\sum_{s \in \mathcal{S}} (\hat{\theta}_{AI}(s) - \hat{\theta}_{human}(s)). \tag{46}$$

By the Central Limit Theorem, $\hat{\mathbf{b}}_A \sim (\mathbf{b}_A - \mathbf{b}_H, (\sigma_A^2/M_A + \sigma_H^2/M_H)/|\mathcal{S}|)$. Testing $H_0 : \mathbf{b}_A = \mathbf{b}_H$ requires $|\mathcal{S}| \geq z_{\alpha/2}^2(\sigma_A^2/M_A + \sigma_H^2/M_H)/\delta^2$ for power $1 - \beta$ at effect size δ.

Corollary 7.4 (Optimal AI-Human Grader Allocation AIGrader)**.** *Given a budget C with human grader cost c_H and AI grader cost $c_A \ll c_H$, the variance-minimizing allocation solves:*

$$\min_{M_H, M_A} \frac{\sigma_H^2}{M_H} + \frac{\sigma_A^2}{M_A} \quad s.t. \quad c_H M_H + c_A M_A \leq C. \tag{47}$$

The optimal ratio is:

$$\frac{M_A^*}{M_H^*} = \frac{\sigma_A}{\sigma_H}\sqrt{\frac{c_H}{c_A}}. \tag{48}$$

When AI is 100× cheaper ($c_H/c_A = 100$) but 2× noisier ($\sigma_A/\sigma_H = 2$), the optimal ratio is $M_A^*/M_H^* = 2 \cdot 10 = 20$: deploy 20 AI graders per human grader.

Remark 7.5 (The AI Auditor Paradox AI)*.* AI grading is simultaneously (a) the cheapest way to achieve high M (reducing σ^2/M through multiplicity) and (b) the modality most vulnerable to undetected systematic bias (because all AI instances share the same training distribution, creating hidden correlation $\bar{\rho}_{\mathrm{AI}}$). The solution is not to avoid AI grading but to treat AI as one modality in a multi-modal panel that includes human graders. Theorem 7.2 provides the mathematical basis: human graders detect AI bias, while AI graders provide the multiplicity needed for Theorem 3.1's error detection bound.

8 The Cercis Score for Assessment Systems

Definition 8.1 (Cercis Score for Assessment Cercis)**.** For an assessment system $\mathcal{A}$, the Cercis Score is:

$$\textsc{Cercis}(\mathcal{A}) = Q(\mathcal{A}) + \eta \cdot N(\mathcal{A}), \tag{49}$$

where:

- $Q(\mathcal{A}) \in [0, 1]$ is the **Quality Guarantee Quality Assurance**: the fraction of SCX theorems the system structurally satisfies:

$$Q(\mathcal{A}) = \frac{1}{5}\Big[q_1(\mathcal{A}) + q_2(\mathcal{A}) + q_3(\mathcal{A}) + q_4(\mathcal{A}) + q_5(\mathcal{A})\Big], \tag{50}$$

  where:
  - $q_1 \in [0, 1]$: degree to which Theorem 1 (multi-grader detection) is satisfied. $q_1 = 1$ when $M \geq M_{\min}$ as defined in equation (16).
  - $q_2 \in [0, 1]$: degree to which Theorem 2 (self-audit prohibition) is satisfied. $q_2 = 1$ when no grader evaluates their own prior scores; $q_2 = 0$ when the system self-certifies.
  - $q_3 \in [0, 1]$: degree to which Theorem 3 (noise-signal unidentifiability) is addressed. $q_3 = 1$ when rubric dimension contributions are independently auditable.
  - $q_4 \in [0, 1]$: degree to which Situs-1 (encoding quality) is satisfied. $q_4 = 1$ when the rubric has bounded Lipschitz constant and measurable reconstruction error.
  - $q_5 \in [0, 1]$: degree to which Spring-1 (permanent memory) is satisfied. $q_5 = 1$ when all scores, grader identities, and rubrics are permanently stored.
- $N(\mathcal{A}) \in [0, 1]$ is the **Empirical Novelty** : normalized scoring reliability metrics (inter-rater reliability, test-retest stability, predictive validity).
- $\eta = 0.2$ is the **epistemic discount factor** : empirical reliability without formal certification is worth at most 20%.

Critical: If $Q(\mathcal{A}) = 0$, then $\textsc{Cercis}(\mathcal{A}) \leq 0.20$ regardless of empirical reliability. A system with 0.95 inter-rater reliability and zero quality guarantee is epistemically equivalent to random grading with a lucky rubric.

The Cercis Score exposes a systematic pattern: current assessment systems cluster in the 0.08–0.56 range, far below the certification threshold of 0.70. Even the Gaokao's double-grading

Table 2: Cercis Scores for Representative Assessment Systems

System	q_1	q_2	q_3	q_4	q_5	Q	CERCIS
Single teacher, no rubric	0.0	0.0	0.0	0.0	0.1	0.02	0.08
Single teacher + rubric	0.0	0.0	0.2	0.5	0.1	0.16	0.26
Double-blind (M=2)	0.3	0.7	0.3	0.5	0.3	0.42	0.52
Gaokao essay (double + arb.)	0.4	0.7	0.4	0.5	0.3	0.46	0.56
Multi-grader (M=5), blind	0.8	0.9	0.6	0.7	0.5	0.70	0.76
AI only (M_A=50)	0.3	0.0	0.1	0.4	0.8	0.32	0.54
AI + Human hybrid (M=11)	0.9	0.9	0.7	0.8	0.9	0.84	0.88
Full SCX ($M \geq M_{\min}$, SPRING, YAJIE)	1.0	1.0	1.0	1.0	1.0	1.00	1.00

with arbitration — among the most sophisticated operational systems — achieves only CERCIS = 0.56. The gap between current practice and SCX-compliant assessment (CERCIS = 1.00) is the education audit gap quantified.

9 Experimental Protocols

9.1 Protocol 1: Gaokao Essay Scoring Audit

Setting. Gaokao Chinese essay (, 60 points). $N = 10,000$ essays sampled from a provincial examination cohort. Standard protocol: $M = 2$ human raters, third rater if $|v_1 - v_2| > 6$. **SCX Enhancement.**

Step 1: **Grader pool expansion Grader.** Recruit $M = 20$ independent graders from different provinces, trained on disjoint calibration sets (A1 enforcement).

Step 2: **Full blindness .** Redact all student identifiers (name, school, region) and randomize essay order per grader (A1 enforcement, $\beta \to 1$).

Step 3: **AI grader addition AIGrader.** Deploy $M_A = 5$ AI graders (GPT-4, Claude, Qwen, ERNIE, Gemini) with distinct prompts and temperature settings.

Step 4: **YAJIE consensus computation YAJIE.** Compute $\hat{\theta}_{\text{YAJIE}}(s)$ with inverse-variance weights estimated from calibration essays.

Step 5: **SPRING permanent record SPRING.** Store: essay text hash, all $M_H + M_A$ scores, grader identities, rubric version, timestamp, and $\hat{\theta}_{\text{YAJIE}}(s)$ with 95% CI.

Step 6: **Inflation monitoring Inflation.** Compare $\hat{\theta}_{\text{YAJIE}}$ across years; flag when $\iota_t > 0$ with $p < 0.05$.

Expected Outcomes.

- Detection probability for $\Delta = 6$ (10% of 60): $\mathbb{P}(\text{detect}) \geq 1 - \exp(-2 \cdot 20 \cdot 0.01/(1+0.3)) \approx 1 - e^{-0.308} \approx 26.5\%$, versus $\approx 3.9\%$ for $M = 2$.
- AI-human bias estimate $\hat{\mathbf{b}}_A$ with 95% CI width ≤ 3 points for $N = 10,000$.
- CERCIS Score improvement from 0.56 (current) to ≈ 0.85.

9.2 Protocol 2: SAT Essay Scoring Audit SAT

Setting. SAT Essay (discontinued 2021, but illustrative). Scoring: Reading, Analysis, Writing dimensions, each 2–8 points. Two human raters per dimension. **SCX Enhancement.**

Step 1: **Multi-modal panel .** $M_H = 5$ human raters $+$ $M_A = 10$ AI raters across 3 rubric dimensions $\times$ 2 modalities $=$ 30 scores per essay.

Step 2: **Dimension-specific YAJIE consensus Yajie.** Compute $\hat{\theta}^{(k)}_{\text{YAJIE}}(s)$ per dimension k, then aggregate via rubric-weighted sum.

Step 3: **Bias decomposition .** Decompose total grader bias into dimension-specific components: $\mathbf{b}_m = (\mathbf{b}_m^{(1)}, \mathbf{b}_m^{(2)}, \mathbf{b}_m^{(3)})$.

Step 4: **SPRING gating for rubric drift SPRINGRubric.** Detect when grader score distributions shift, indicating rubric interpretation drift or inflation.

9.3 Protocol 3: AI-Assisted Classroom Grading AI

Setting. University course with $N = 200$ students, 4 essay assignments. Instructor grades all essays (traditional $M = 1$). **SCX Enhancement.**

Step 1: **AI pre-grading AI.** Deploy $M_A = 5$ AI graders on all essays ($5 \times 200 \times 4 = 4000$ scores at near-zero marginal cost).

Step 2: **Human spot-check .** Instructor grades a random sample of $n = 40$ essays (10%). This provides calibration for $\mathbf{b}_A$ estimation.

Step 3: **Disagreement flagging Disagreement.** Essays where $|\hat{\theta}_{\text{AI}} - \hat{\theta}_{\text{human}}| > 2\sigma_{\text{disagree}}$ are flagged for additional human review.

Step 4: **CERCIS Score reporting CERCIS.** Each student receives $\hat{\theta}_{\text{YAJIE}}$ with 95% CI, grader composition, and overall CERCIS($\mathcal{A}$) for the course.

Cost-Benefit. Human grading cost: $200\times4\times15$min $=$ 200 hours. SCX cost: $40\times4\times15$min $=$ 40 hours + AI API costs (∼\$50). Net savings: 160 hours (80%), with formal certification.

10 Discussion

10.1 The Cost of Certification

The primary objection to multi-grader SCX audit is cost: M graders cost M times as much as one. We address this at three levels:

(i) **AI reduces marginal cost AI.** AI graders cost ∼\$0.01–\$0.10 per essay versus ∼\$5–\$20 for human graders (at ∼\$20–\$80/hour, 4 essays/hour). The optimal AI:human ratio from Corollary 7.4 shows that even modest AI deployment ($M_A = 10$–50) dramatically increases M_{eff} at minimal cost.

(ii) **Not all assessments need full certification .** Formative classroom assessment may accept CERCIS $\approx$ 0.3–0.5; high-stakes examinations require CERCIS ≥ 0.8. The CERCIS Score enables cost-graded certification: higher stakes $\rightarrow$ higher required CERCIS $\rightarrow$ higher M.

(iii) **The cost of *not* certifying .** A single erroneous Gaokao score can alter a student's university trajectory, with estimated lifetime earnings impact of \$50,000–\$200,000. At 12M essays/year with $M = 2$ and $\mathbb{P}(\text{miss } \Delta = 10\%) \approx 97.4\%$, the expected number of undetected significant errors is large. The certification cost ($M = 10$ graders $\times$ \$50/essay $=$ \$500/essay incremental) must be weighed against the cost of errors.

10.2 Irreducible Inter-Grader Correlation

A2 (conditional independence) is an idealization. In practice, graders share:

- **The same rubric $\mathcal{R}$:** even with independent training, the rubric constrains all graders toward the same score regions. This is desirable (it reduces variance) but creates irreducible correlation.

- **The same cultural context** : graders from similar educational backgrounds share implicit scoring norms not captured by the rubric.
- **The same language** : for essay scoring, linguistic patterns that one grader finds sophisticated, another also finds sophisticated.

The framework accommodates this through $M_{\text{eff}} = M/(1 + \bar{\rho})$. The irreducible correlation $\bar{\rho}_{\min}$ is an empirical quantity that must be estimated for each assessment context. The Cercis Score should report $\bar{\rho}_{\min}$ alongside M_{eff}.

10.3 Rubric Incommensurability Rubric

Theorem 4.1 assumes a common rubric $\mathcal{R}$ shared by all graders. But what if graders use *different* rubrics? Consider:

- **Holistic vs. analytic vs. .** Grader A uses a holistic 1–6 scale; Grader B uses analytic scoring across 5 dimensions. Their scores are incommensurable — they cannot be averaged.
- **Criterion disagreement Disagreement.** Grader A values "creativity"; Grader B values "grammatical accuracy." Their scores reflect different latent constructs.

SCX addresses this through Situs encoding: each grader's rubric $\mathcal{R}_m$ defines a distinct encoding $\phi_m :\to [0, B_m]$. The Yajie consensus operates not on raw scores but on the *rankings* induced by each rubric: $\hat{\theta}_{\text{Yajie}} = \text{median}_m(\text{rank}_m(w_s))$. This preserves the consensus property while accommodating rubric heterogeneity. The cost is reduced interpretability — a consensus rank is less informative than a consensus score.

10.4 The Political Economy of Grade Inflation ScoreInflation

Theorem 5.2 proves that grade inflation is detectable with temporally-independent grader pools. But detection is not correction. Grade inflation persists because institutions benefit from it:

- **Student satisfaction .** Higher grades increase student satisfaction and reduce complaints.
- **Competitive positioning .** Institutions with higher average grades appear more successful.
- **Graduate outcomes .** Higher grades improve graduate school and employment prospects, creating alumni goodwill.

SCX does not solve the political economy problem — it solves the detection problem. Once inflation is detected and publicly certified (via Spring permanent records), the political choice to tolerate or correct it becomes explicit rather than hidden. This is the governance analog: audit creates transparency; transparency enables accountability; accountability is a political choice.

10.5 Limitations

We declare the following honest limitations:

[L1] **Latent score unobservability Score.** The true score θ_s is a latent construct. All theorems are stated in terms of θ_s, but θ_s is never directly observed. The framework certifies *consensus* — agreement among graders — not *truth*. Consensus is a necessary condition for truth (disagreement implies at least one grader is wrong) but not sufficient (all graders could share the same bias).

[L2] **AI grader correlation AIGrader.** AI graders from different providers (GPT-4, Claude, Gemini) may have correlated errors because they are trained on overlapping internet text corpora. The independence assumption A2 for AI graders requires empirical validation. If $\bar{\rho}_{\text{AI}} \approx 0.3$–$0.5$, then deploying $M_A = 50$ AI graders is equivalent to $M_{\text{eff}} \approx 33$–$38$, not 50.

[L3] **Rubric quality is exogenous Rubric.** The framework certifies scoring *given* a rubric, not the rubric itself. A poor rubric ($\varepsilon_{\mathcal{R}}$ large, $L_{\mathcal{R}}$ large) can produce perfectly certified worthless scores. Rubric validation — demonstrating that rubric scores predict external outcomes — is outside the scope of SCX audit.

[L4] **Cost remains a barrier .** While AI reduces marginal cost, human graders are still required for bias calibration (Theorem 7.2). The minimum viable M_H is likely 3–5, which may be prohibitive for resource-constrained settings.

[L5] **Temporal grader drift Grader.** Individual graders' biases $\mathbf{b}_m(t)$ may drift over time (fatigue, experience, rubric re-interpretation). SPRING memory enables detection of this drift, but correction requires re-calibration, which may lag behind the drift.

[L6] **Cultural specificity .** The rubric consensus in Theorem 4.1 assumes graders share a common understanding of quality. In cross-cultural assessment (e.g., PISA, TOEFL), graders from different cultures may apply incompatible quality standards. SCX detects the resulting disagreement but cannot resolve it — resolution requires substantive consensus on what quality means, which is a normative question.

11 Conclusion Conclusion

Educational assessment is the world's largest measurement enterprise, and it operates without formal certification. Scores are claims about student competence; graders are experts making those claims. The SCX auditing framework, applied to this domain, reveals that:

(i) **Single-grader scoring is epistemically equivalent to no scoring** ($M = 1 \Rightarrow Q = 0$). A teacher, examiner, or AI that scores alone cannot certify the score's accuracy. This is not a criticism of grader competence — it is a mathematical consequence of Theorem 2 (self-audit = no audit).

(ii) **Current "best practices" (Gaokao double-grading, SAT two-rater) provide negligible error detection** ($\mathbb{P}(\text{detect } \Delta = 10\%) \leq 3\%$–$5\%$). Two correlated graders are not meaningfully better than one.

(iii) **AI grading is not a threat — it is the enabling technology for affordable multi-grader audit.** AI graders can provide $M_A = 50$–100 at near-zero marginal cost, while $M_H = 3$–5 human graders provide bias calibration. The hybrid model achieves CERCIS > 0.85 at lower total cost than current $M = 2$ human-only systems.

(iv) **Grade inflation is mathematically detectable** but requires temporally-independent grader pools and SPRING permanent records. The framework transforms inflation from an anecdotal concern into a testable hypothesis with explicit detection bounds.

(v) **The CERCIS Score provides a single-number summary** of assessment system certification quality, enabling comparison across systems, institutions, and countries. Any assessment system can compute and publish its CERCIS Score — and should be asked to. The bottom line: **every score is a claim. Every claim needs an auditor. In education, the auditors are other graders. The more independent graders agree, the more the score can be trusted.** This is not a pedagogical opinion — it is a mathematical theorem. **Conclusion.** Educational Assessment, .Score; Grader.SCX, : (1)Graderawaiting; (2)"", SAT; (3)AI——Grader; (4)ScoreInflation, GraderSPRING; (5)CERCIS.: Score.., Grader.Grader, Score.——.

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